Kruskal-Wallis Test

2026-04-17

Introduction

The Kruskal-Wallis test extends the Mann-Whitney U to three or more independent groups. It tests whether the distributions differ across groups, without assuming normality or equal variances. Post-hoc pairwise comparisons use Dunn's test with a multiple-testing correction.

Prerequisites

Mann-Whitney U, ranks.

Theory

Pool all observations and rank them across groups. Let R_i be the sum of ranks in group i with size n_i, and $N = \sum n_i$. The Kruskal-Wallis statistic is

$$H = \frac{12}{N(N+1)} \sum_i \frac{R_i^2}{n_i} - 3(N+1) \sim \chi^2_{k-1}.$$

Under H_0 of equal distributions, H has an approximate chi-squared distribution with $k-1$ df for moderate sample sizes.

Effect size: $\varepsilon^2 = H/(N-1)$ or $\eta^2_H = (H-k+1)/(N-k)$.

Post-hoc: Dunn's test compares each pair of groups using z-statistics with Bonferroni or Benjamini-Hochberg correction.

Assumptions

- Independent observations within and across groups.
- Observations at least ordinal.
- Similar distribution shapes (for a median-difference interpretation); otherwise tests distributional equality.

R Implementation

```r
library(rstatix); library(dunn.test)
set.seed(2026)

df <- data.frame(
  group = factor(rep(c("A", "B", "C", "D"), each = 25)),
  y     = c(rnorm(25, 50, 8),
            rnorm(25, 55, 8),
            rnorm(25, 60, 12),
            rnorm(25, 52, 8))
)
```

```r
kruskal.test(y ~ group, data = df)

# Effect size
df |> kruskal_effsize(y ~ group)

# Dunn's post-hoc with BH correction
dunn.test(df$y, df$group, method = "bh")
```

Output & Results

```
        Kruskal-Wallis rank sum test
Kruskal-Wallis chi-squared = 15.22, df = 3, p-value = 0.0016

Cohen's eta2 | 95% CI
------------------------
0.14          | [0.02, 1.00]

Dunn's test with Benjamini-Hochberg adjustment:
     Comparison    p.adj
     A - B         0.086
     A - C         0.003
     A - D         0.302
     B - C         0.136
     B - D         0.302
     C - D         0.015
```

Interpretation

"Group differed significantly across the four groups (Kruskal-Wallis $H(3) = 15.22$, $p = 0.002$, $\eta^2_H = 0.14$). Dunn post-hoc tests identified group C as significantly higher than groups A and D (BH-adjusted $p < 0.05$)."

Practical Tips

- Report medians and IQRs per group.
- Dunn's test is the standard post-hoc procedure; pairwise Wilcoxon with adjustment is an alternative.
- For within-subjects designs, use Friedman.
- When the only question is whether any groups differ, skipping the omnibus test and doing pairwise Wilcoxons with adjustment is statistically valid but loses interpretability.
- For heavy tails or strong shape differences, the test compares distributions more broadly than medians.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests

- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests